SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: COMPUTER VISION with Open CV

COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO3: Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)

Assessment Tool Weightage: 20%

Read the given scenario carefully and analyze the problem using appropriate computer vision techniques. Justify your answers with relevant reasoning and comparisons wherever necessary.

Code of Conduct:

I *N HARSHAVARDHAN (192324189)* certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N HARSHAVARDHAN

ANSWERS AND SOLUTIONS

1. Preprocessing and Feature Detection

Scenario:

A vision system processes images captured under varying lighting conditions. Without preprocessing, feature detection results are inconsistent. Analyze the issue and explain how preprocessing improves feature detection performance.

Analysis:

- Varying illumination changes pixel intensity values across images of the same scene, even though the underlying structure is unchanged.
- Feature detectors such as Harris corner detector, SIFT, and ORB rely on local intensity gradients. When lighting shifts (brightness, shadows, glare), gradient magnitudes and directions become inconsistent between frames.
- This causes the same physical keypoint to be detected differently (or missed entirely) across images, reducing repeatability - the core requirement for reliable feature detection and matching.
- Low-contrast or noisy regions caused by poor lighting can also generate spurious keypoints (false positives) or suppress genuine ones (false negatives).

Solution / Improvement:

- Grayscale conversion and intensity normalization: removes color-channel bias and rescales intensity range for consistency.
- Histogram Equalization / CLAHE (Contrast Limited Adaptive Histogram Equalization): redistributes intensity values to enhance local contrast without over-amplifying noise, making features stand out uniformly across lit and shadowed regions.
- Gaussian smoothing: reduces high-frequency sensor noise before gradient computation, stabilizing detector response.
- Illumination normalization techniques (e.g., homomorphic filtering, Retinex) separate reflectance from illumination, making features invariant to lighting changes.
- Gamma correction: compensates for non-linear sensor response under bright/dark conditions.

Conclusion:

Preprocessing standardizes the intensity distribution of the image before feature detection, which increases the repeatability and stability of detected keypoints across varying lighting conditions, directly improving the accuracy of downstream matching tasks.

2. Edge vs Corner Detection

Scenario:

In an image containing both straight edges and textured regions, a system uses edge detection but fails to identify key points for matching. Analyze why edge detection is insufficient and justify the use of corner detection in this case.

Analysis:

- Edge detectors (e.g., Canny, Sobel) identify pixels where intensity changes sharply in one direction, producing continuous curves along object boundaries.
- An edge point, by itself, is ambiguous for matching - it suffers from the 'aperture problem': along a straight edge, every point looks locally identical, so a matching algorithm cannot

uniquely localize corresponding points between two images.

- Edges provide good shape/boundary information but poor point-to-point correspondence because they lack distinctiveness along their length.

Solution / Improvement:

- Corners (points where intensity changes significantly in two orthogonal directions) are far more distinctive and can be localized precisely in both x and y directions.
- Detectors like Harris Corner Detector or Shi-Tomasi (Good Features to Track) evaluate the local structure using the second-moment matrix (gradient covariance), identifying points with high curvature/intensity variation in all directions.
- Because corners are unique in their local neighborhood, they satisfy the matching requirement of distinctiveness and repeatability, making them suitable as keypoints for correspondence, tracking, and homography estimation.
- In textured regions, corner detection naturally produces many strong keypoints since texture creates multi-directional gradients.

Conclusion:

Edge detection is insufficient for matching because it lacks point-wise distinctiveness, whereas corner detection identifies uniquely localizable points, making it the appropriate technique for feature matching in images with both straight edges and texture.

3. Hough Transform in Noisy Images

Scenario:

A system uses Hough Transform to detect lines in noisy images but produces inaccurate results. Analyze the effect of noise on detection and suggest improvements based on preprocessing or parameter selection.

Analysis:

- The Hough Transform detects lines by mapping edge points from image space to a parameter space (ρ, θ) and finding accumulator cells with high votes.
- Noise introduces spurious edge pixels that do not belong to actual lines. Each noisy pixel casts votes in the accumulator array, creating false peaks or spreading votes so genuine line peaks become less prominent.
- Noise also causes broken/fragmented edges, which lowers the vote count for true lines, making them harder to distinguish from noise-induced peaks.
- An improperly chosen accumulator resolution (bin size for ρ and θ) or threshold can worsen this - too fine a resolution splits votes for a true line across multiple bins; too coarse merges distinct lines.

Solution / Improvement:

- Apply Gaussian blur before edge detection to suppress high-frequency noise. • Use Canny edge detection (with proper hysteresis thresholds) instead of raw gradient thresholding, since Canny already includes noise suppression and non-maximum suppression, producing thinner and cleaner edges as Hough input.
- Tune the accumulator bin size for ρ and θ appropriately - coarser bins increase noise tolerance while finer bins improve precision; a balance avoids both missed detections and false merges. • Increase the voting threshold so only accumulator cells with sufficiently strong support (many collinear edge points) are accepted as lines, filtering out weak noise-driven peaks. • Use the

Probabilistic Hough Transform, which also enforces minimum line length and maximum line gap parameters to reject short, noise-generated segments.

Conclusion:

Noise degrades Hough Transform accuracy by generating false votes and fragmenting true edges. Combining noise-reduction preprocessing (smoothing, robust edge detection) with careful parameter tuning (bin size, threshold, min-length/gap) restores reliable line detection.

4. Feature Matching under Transformations

Scenario:

Two images of the same object are taken at different scales and orientations. Traditional feature matching fails, but scale invariant feature matching works effectively. Analyze why scale invariant methods perform better in this scenario.

Analysis:

- Traditional corner-based detectors (e.g., basic Harris) operate at a single, fixed scale. When the object size changes between images (zoom, distance change), the same physical feature no longer produces the same response, so keypoints detected in one image may not exist at the corresponding location/scale in the other.
- Similarly, fixed-size descriptors (raw pixel patches) are sensitive to rotation - a patch computed at 0 degrees will not match the same patch rotated by 45 degrees, since pixel intensity ordering changes.
- This mismatch in scale and orientation causes traditional matching (based on direct patch/pixel comparison) to fail.

Solution / Improvement:

- Scale-Invariant Feature Transform (SIFT) builds a scale-space pyramid using Difference of Gaussians (DoG) across multiple octaves and scales, detecting keypoints that are stable extrema across scales - ensuring the same physical point is found regardless of image scale.
- Each keypoint is assigned a dominant orientation based on local gradient directions; the descriptor is then computed relative to this orientation, making it rotation invariant. • The descriptor itself (a 128-dimensional histogram of gradient orientations) is normalized for contrast, giving additional robustness to illumination change.
- Alternatives such as SURF and ORB (using FAST + BRIEF with orientation compensation) achieve similar scale/rotation invariance with lower computational cost.

Conclusion:

Scale-invariant methods perform better because they detect and describe keypoints relative to an automatically estimated characteristic scale and orientation, so the resulting descriptors remain consistent under zoom and rotation, enabling correct correspondence between the two images.

5. PCA for Feature Optimization

Scenario:

A system extracts a large number of features, leading to high computational cost and redundancy. Analyze how Principal Component Analysis helps in reducing dimensionality and improving efficiency.

Analysis:

- High-dimensional feature vectors (e.g., large SIFT descriptor sets, raw pixel intensities) often contain correlated/redundant information - many dimensions carry overlapping or low-variance information that does not contribute meaningfully to discrimination.

- This redundancy increases storage, computation time for matching/classification, and can even hurt performance due to the curse of dimensionality (sparse data in high-dimensional space, overfitting in learning models).

Solution / Improvement:

- PCA identifies the directions (principal components) in feature space along which the data varies the most, computed via eigen-decomposition of the covariance matrix (or SVD of the data matrix).
- By projecting the original feature vectors onto the top-k principal components (those with the largest eigenvalues), PCA retains the maximum possible variance (information) while reducing the number of dimensions from the original d to a much smaller k.
- Because principal components are orthogonal/uncorrelated, PCA also removes redundancy between features.
- The reduced feature set requires less memory and significantly speeds up subsequent operations such as nearest-neighbor matching, classification, or clustering.

Conclusion:

PCA improves efficiency by compressing high-dimensional, redundant feature vectors into a compact, decorrelated representation that preserves most of the discriminative information, reducing computational cost without significant loss of accuracy.

6. Preprocessing Impact on Edge Detection

Scenario:

A system performs edge detection on raw images containing noise and uneven illumination, resulting in fragmented edges. Analyze the problem and explain how preprocessing improves edge detection accuracy and continuity.

Analysis:

- Edge detectors compute image gradients; noise creates spurious local intensity fluctuations that produce false gradient responses, leading to broken or discontinuous edge segments and extra false-positive edges.
- Uneven illumination causes gradual intensity changes across the image (e.g., shadows), which can either mask true edges (low contrast in dark regions) or create false edges at illumination boundaries.

Solution / Improvement:

- Apply Gaussian smoothing to suppress high-frequency noise before gradient computation, reducing false edge fragments.
- Use adaptive/local thresholding or CLAHE to normalize contrast across regions with uneven lighting, so true edges are equally detectable everywhere in the image.
- Apply the Canny edge detector, which integrates Gaussian smoothing, gradient computation, non-maximum suppression, and hysteresis thresholding (dual thresholds) - hysteresis in particular reconnects weak edge pixels that are connected to strong ones, improving continuity. • Morphological operations (dilation followed by erosion / closing) can be applied

post-detection to bridge small gaps in fragmented edges.

Conclusion:

Preprocessing (smoothing + contrast normalization) combined with a noise-robust detector such as Canny with hysteresis thresholding significantly improves edge continuity and accuracy under noisy, unevenly lit conditions.

7. Corner Detection vs Edge Detection in Tracking

Scenario:

A tracking system uses edge detection but fails to track moving objects accurately across frames. Analyze why edges are insufficient and justify the use of corner detection for reliable tracking.

Analysis:

- Object tracking requires establishing precise point correspondences between consecutive frames (optical flow / feature tracking).
- Edge points suffer from the aperture problem - along a straight edge, motion can only be estimated perpendicular to the edge direction, not along it, so the true displacement vector is ambiguous.
- This ambiguity leads to drift and incorrect correspondence when only edges are used, especially when the object or camera moves.

Solution / Improvement:

- Corners provide two-dimensional intensity variation, giving a well-defined, unique local signature that can be tracked unambiguously in both x and y directions - satisfying the aperture problem-free condition required by optical flow methods (e.g., Lucas-Kanade).
- The Shi-Tomasi 'Good Features to Track' criterion explicitly selects corners with strong eigenvalues in the structure tensor, which are proven to be optimal for tracking stability. • Corner-based keypoints are more repeatable across frames, allowing robust frame-to-frame association even under moderate motion, rotation, or partial occlusion.

Conclusion:

Because corners resolve the aperture problem and offer distinctive, well-localized points, corner detection (e.g., Shi-Tomasi/Harris combined with KLT tracking) is justified over pure edge detection for reliable multi-frame object tracking.

8. Hough Transform for Shape Detection

Scenario:

A system uses Hough Transform to detect circular shapes but fails in complex backgrounds. Analyze the limitations and explain how preprocessing or parameter tuning can improve detection.

Analysis:

- The circular Hough Transform searches a 3D parameter space (center x, center y, radius), which is far more computationally expensive and sensitive to noise than the 2D line detection case.
- In complex/cluttered backgrounds, many unrelated edges (textures, other objects) generate spurious votes in the accumulator, producing false circle detections or masking true ones. • If

the radius range is not properly constrained, the search space explodes and false positives increase; overlapping or partially occluded circles also reduce vote concentration, weakening true peaks.

Solution / Improvement:

- Preprocess with Gaussian blur and Canny edge detection to retain only strong, clean edges, reducing spurious votes from background clutter.
- Restrict the radius search range (minRadius, maxRadius) to plausible values based on prior knowledge of the object size, shrinking the search space and reducing false positives. • Use gradient-based methods (e.g., OpenCV's HOUGH_GRADIENT) which use edge gradient direction to vote more selectively, rather than blind voting by every edge pixel. • Tune accumulator threshold (param2) and minimum distance between circle centers to suppress overlapping/duplicate false detections.

- Apply background subtraction or region-of-interest (ROI) masking, if the object location is roughly known, to eliminate irrelevant clutter before running the transform.

Conclusion:

Complex backgrounds overwhelm the circular Hough Transform's voting process with irrelevant edges. Careful preprocessing (denoising, clean edges, ROI restriction) combined with tuned parameters (radius range, accumulator threshold, minimum center distance) substantially improves detection reliability.

9. Feature Matching under Illumination Changes

Scenario:

Two images of the same scene are captured under different lighting conditions, causing feature matching errors. Analyze why matching fails and explain how preprocessing or robust descriptors improve performance.

Analysis:

- Illumination changes alter the absolute intensity values and contrast of corresponding regions between the two images, even though the scene structure is unchanged.
- Descriptors that rely directly on raw pixel intensity (without normalization) produce different values for the same physical point under different lighting, causing false matches or missed correspondences.
- Shadows and highlights can also change the local gradient pattern, further degrading matching for gradient-sensitive descriptors.

Solution / Improvement:

- Apply histogram equalization / CLAHE to both images to normalize contrast and reduce illumination discrepancy before feature extraction.
- Use descriptors that are inherently robust to illumination, such as SIFT, which normalizes its gradient-orientation histogram by vector length, making it invariant to uniform (affine) intensity changes.
- Binary descriptors like ORB/BRIEF compare relative pixel intensities rather than absolute values, providing some robustness to monotonic illumination changes. • Apply illumination normalization techniques (e.g., Retinex algorithm, homomorphic filtering) to separate lighting effects from actual scene reflectance before matching.
- Use ratio-based matching tests (e.g., Lowe's ratio test) to filter out ambiguous matches that arise from illumination-induced noise.

Conclusion:

Matching fails because raw intensities/gradients change with illumination, but combining illumination-normalizing preprocessing with descriptors designed for photometric invariance (SIFT, ORB) restores reliable correspondence across differently lit images.

10. PCA in Face Recognition

Scenario:

A face recognition system processes high-dimensional image data, resulting in slow performance. Analyze how PCA helps in reducing dimensionality while preserving important features.

Analysis:

- A face image of size, e.g., 100x100 pixels represents a 10,000-dimensional vector. Processing, storing, and comparing such high-dimensional vectors across a large database is computationally expensive and prone to the curse of dimensionality.
- Much of this pixel-level information is redundant, since neighboring pixels and facial structure are highly correlated across the dataset.

Solution / Improvement:

- The classic 'Eigenfaces' approach applies PCA to a training set of face images: it computes the covariance matrix of the mean-centered face vectors and extracts the eigenvectors (eigenfaces) corresponding to the largest eigenvalues.
- Each face can then be represented as a weighted combination of a small number (k) of eigenfaces instead of the full pixel vector, drastically reducing dimensionality (e.g., from 10,000 to 50-150 dimensions) while retaining the components that capture the most variance (the most visually significant facial features).
- Recognition/matching is then performed in this compact 'face space' using distance metrics (e.g., Euclidean distance) between projected coefficients, which is far faster than comparing raw images.
- Selecting k based on cumulative explained variance (e.g., retaining 95% of variance) balances speed and recognition accuracy.

Conclusion:

PCA (Eigenfaces) reduces the high-dimensional face image space to a compact set of principal components that capture the dominant facial variation, enabling much faster processing and comparison while preserving the features most important for recognition.

11. Edge Detection in Low Contrast Images

Scenario:

An image has low contrast, causing edge detection algorithms to fail. Analyze the problem and explain how preprocessing enhances edge detection results.

Analysis:

- Edge detectors identify boundaries based on the magnitude of intensity gradients. In low contrast images, the intensity difference between objects and background (or between adjacent regions) is small, so gradient magnitudes fall below typical detection thresholds.
- This causes real edges to be missed (false negatives), and the detector may instead respond

to noise, whose relative magnitude becomes more significant when overall contrast is low.

Solution / Improvement:

- Apply histogram equalization to redistribute intensity values across the full dynamic range, stretching contrast between regions that were previously close in intensity.
- Use CLAHE (adaptive histogram equalization) for localized contrast enhancement without over-amplifying noise in already-bright or already-dark regions.
- Apply contrast stretching / gamma correction to expand the effective intensity range before gradient computation.
- Lower or adaptively tune the gradient threshold in the edge detector (e.g., Canny's low/high thresholds) to accommodate the smaller gradient magnitudes typical of a low-contrast (but now enhanced) image.
- Combine with mild denoising, since contrast enhancement can also amplify existing noise.

Conclusion:

Low contrast weakens gradient responses needed for edge detection. Contrast-enhancing preprocessing (histogram equalization/CLAHE, gamma correction) restores sufficient gradient magnitude for edges to be reliably detected.

12. Corner Detection in Textured Regions

Scenario:

A system applies corner detection in highly textured regions, resulting in excessive keypoints. Analyze the issue and suggest methods to improve feature selection.

Analysis:

- Highly textured regions (e.g., foliage, fabric patterns, gravel) contain intensity variation in multiple directions almost everywhere, so corner detectors (Harris, Shi-Tomasi, FAST) respond strongly at a very large number of locations.
- An excessive number of keypoints increases computational cost for description and matching, introduces many weak/unstable keypoints (sensitive to small noise), and can dilute the truly distinctive features among a flood of low-quality ones.

Solution / Improvement:

- Apply a stronger corner-response threshold, keeping only keypoints with the highest 'cornerness' score (e.g., higher minimum eigenvalue in Shi-Tomasi, higher Harris response R).
- Use Non-Maximum Suppression (NMS) within a spatial neighborhood so that only the strongest local keypoint is retained, removing tightly clustered redundant corners.
- Limit the maximum number of keypoints (e.g., goodFeaturesToTrack's maxCorners parameter) and rank by response strength, keeping only the top-N most significant points.
- Enforce a minimum spatial distance between accepted keypoints to ensure a more uniform, less redundant spatial distribution.
- Apply scale-space based detectors (SIFT/SURF) which naturally suppress low-contrast and edge-like weak keypoints via additional filtering steps (e.g., DoG extremum thresholding and edge-response elimination).

Conclusion:

Excessive keypoints in textured regions arise because many points satisfy the basic corner criterion. Thresholding by response strength, non-maximum suppression, and capping the keypoint count select a smaller set of the most stable and distinctive features, improving both efficiency and matching quality.

13. Feature Matching with Noise

Scenario:

Feature matching between two images fails due to noise in one image. Analyze the impact of noise and explain how preprocessing improves matching accuracy.

Analysis:

- Noise perturbs pixel intensities, which alters local gradients and intensity patterns used to compute keypoint locations and descriptors.
- This can shift detected keypoint positions slightly, change descriptor vectors (e.g., gradient histograms in SIFT, intensity comparisons in ORB), or introduce entirely spurious keypoints that have no counterpart in the clean image.
- The result is an increase in false matches (incorrect correspondences) and a decrease in true positive matches, since descriptors from the noisy image no longer align closely with descriptors from the clean image in feature space.

Solution / Improvement:

- Apply Gaussian or median filtering to denoise the affected image before feature detection - median filtering is particularly effective against salt-and-pepper style noise while preserving edges.

- Use descriptors that are inherently more robust to noise, such as SIFT's normalized gradient histograms, which average gradient information over a neighborhood rather than relying on individual pixel values.
- Apply a distance-ratio test (Lowe's ratio test) during matching to reject ambiguous matches likely caused by noise-induced descriptor distortion.
- Use outlier rejection methods like RANSAC when estimating a geometric transform (e.g., homography) between the two images, discarding matches inconsistent with the majority geometric model.

Conclusion:

Noise distorts keypoint locations and descriptors, causing unreliable matches. Denoising preprocessing combined with robust descriptors and geometric outlier rejection (RANSAC, ratio test) restores accurate feature matching.

14. PCA and Information Loss

Scenario:

A system applies PCA aggressively and loses important feature details. Analyze the trade-off between dimensionality reduction and information loss, and justify optimal component selection.

Analysis:

- PCA reduces dimensionality by projecting data onto the top-k principal components, ranked by the variance (eigenvalue) they explain. Choosing too small a k discards components that, while individually smaller in variance, may still carry discriminative information important for the task (e.g., fine facial features, subtle texture differences).
- This creates a fundamental trade-off: more components retained -> more information preserved but less compression/speed gain; fewer components -> greater efficiency but risk of

discarding meaningful variance, degrading downstream accuracy (classification, matching, recognition).

Solution / Improvement:

- Select k based on cumulative explained variance ratio - commonly retaining enough components to preserve 90-99% of total variance, rather than an arbitrary small fixed number.
- Use a scree plot (eigenvalues vs. component index) to identify the 'elbow point' where additional components contribute diminishing returns in variance explained.
- Cross-validate the downstream task performance (e.g., recognition accuracy) for different values of k, selecting the smallest k that keeps performance within an acceptable margin of the full-dimensional baseline.
- Consider that variance is a proxy for importance, not a guarantee - in some cases (e.g., class discriminative tasks) supplementing PCA with methods like Linear Discriminant Analysis (LDA) helps retain task-relevant information that pure variance-based PCA might deprioritize.

Conclusion:

Aggressive PCA loses important detail because it discards components blindly based on variance without considering task relevance. Selecting the number of components based on cumulative explained variance, scree-plot analysis, and validated task performance balances dimensionality reduction with acceptable information retention.

15. Combined Feature Detection and Matching

Scenario:

A system detects features correctly but fails during matching due to incorrect descriptor selection. Analyze the issue and explain how appropriate feature descriptors improve matching performance.

Analysis:

- Feature detection (finding keypoint locations) and feature description (encoding the local appearance around each keypoint) are separate stages. Even with well-detected, stable keypoints, matching can fail if the descriptor used does not adequately capture distinctive, invariant information about the local neighborhood.
- For example, using a simple raw-intensity patch descriptor is sensitive to rotation, scale, and illumination changes; using a low-dimensional or poorly discriminative descriptor increases the chance of false matches between visually similar but distinct keypoints; mismatching descriptor types (e.g., comparing binary descriptors with a Euclidean-distance matcher instead of Hamming distance) also causes systematic matching failure.

Solution / Improvement:

- Choose a descriptor suited to the application's invariance requirements: SIFT/SURF for robustness to scale, rotation, and moderate illumination change (floating-point, gradient histogram based); ORB/BRIEF/BRISK for fast, binary descriptors suited to real-time applications, using Hamming distance for matching.
- Ensure the distance metric used in matching corresponds to the descriptor type - Euclidean/L2 distance for float descriptors (SIFT/SURF), Hamming distance for binary descriptors (ORB/BRIEF).
- Apply Lowe's ratio test to filter ambiguous matches by comparing the distance of the best match to the second-best match.
- Use a robust matcher such as FLANN (Fast Library for Approximate Nearest Neighbors) for large datasets with float descriptors, and Brute-Force-Hamming for binary descriptors. •

Follow matching with geometric verification (RANSAC-based homography/fundamental matrix estimation) to eliminate any remaining outlier matches.

Conclusion:

Correct keypoint detection alone does not guarantee good matching - the descriptor must be chosen to match the required invariances (scale, rotation, illumination) and paired with the correct distance metric and matcher.

Selecting an appropriate descriptor (e.g., SIFT for accuracy, ORB for speed) and validating matches geometrically significantly improves matching performance.

RUBRICS FOR CALCULATION:

Criteria	Weightag	– Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Failsto apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandab le presentation	Limited clarity and structure	Poor clarity; difficult to understand